



Strathmore University

**School of Computing and
Engineering Sciences**



Leveraging LoRaWAN at Strathmore University: A Gateway to IoT Innovation in Research and Education

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Abstract

This article offers guidance on how students, researchers, and faculty members at Strathmore University can access and utilize the campus Long Range Wide Area Network (LoRaWAN) network infrastructure for Internet of Things (IoT) research, educational projects, and innovative applications. The document provides technical specifications and access procedures for leveraging this wireless technology in academic pursuits.

1. Introduction

The Makerspace under the School of Computing and Engineering Sciences (SCES) deployed a LoRaWAN network on the STC building to cover the whole campus and foster innovation in Internet of Things research and education. This network represents a significant technological investment that positions the university at the forefront of IoT development in Africa, providing students and researchers with hands-on access to industry-standard wireless communication technology.

The LoRaWAN network serves as a living laboratory where theoretical knowledge meets practical application, enabling breakthrough research in environmental monitoring, smart agriculture, building automation, and numerous other use cases. By leveraging this infrastructure, the Strathmore community can contribute meaningfully to the global IoT ecosystem while addressing local challenges through innovative technological solutions.

2. Understanding LoRaWAN

LoRaWAN is a Low Power Wide Area Network (LPWAN) protocol that operates on unlicensed Industrial, Scientific, and Medical (ISM) frequency bands. The technology utilizes LoRa modulation, a proprietary chirp spread spectrum (CSS) technique that enables long-range communication with minimal power consumption. This revolutionary approach to wireless communication addresses the specific needs of IoT applications that require long battery life, extensive coverage, and reliable data transmission.

The fundamental principle behind LoRaWAN lies in its ability to transmit small amounts of data over vast distances while consuming minimal power. Unlike traditional wireless technologies that prioritize high data rates, LoRaWAN optimizes for range and efficiency. This technology makes it ideal for applications where devices need to operate unattended for years while maintaining connectivity over several kilometers.

This network implements a sophisticated communication protocol featuring bidirectional communication with both uplink and downlink message capability. Adaptive Data Rate (ADR) provides automatic optimization of transmission parameters based on link quality and network conditions. End-to-end security utilizes AES-128 encryption for all data transmissions, protecting sensitive research data. Message acknowledgment ensures reliable delivery confirmation for critical data applications. Over-the-air updates enable remote firmware and configuration management, reducing maintenance overhead for large deployments.

Technical Characteristics

Parameter	Specification
Frequency Band	863-870 MHz (EU868 plan)
Modulation	Chirp Spread Spectrum (CSS)
Range	Up to 20 km rural, 3-7 km urban
Data Rates	0.3 kbps to 50 kbps
Security	AES-128 encryption

3. Network Coverage Capabilities

The LoRaWAN network is deployed at Strathmore University's topmost point with the aim of providing coverage across the entire campus to support IoT applications in both indoor and outdoor environments. Soon, a detailed coverage test will be done to assess the need for smaller gateways that can offer redundancy.

4. Eligibility and Access Policy

Prohibited Activities are **NOT ALLOWED** to maintain network integrity and academic focus. Unauthorized commercial applications that bypass university oversight are strictly forbidden. Network security testing or penetration attempts that could compromise network security are prohibited without **explicit authorization**. Interference with other users' devices or data violates community standards and

academic ethics. Activities violating university IT policies or local regulations are subject to disciplinary action.

Permitted Activities

User Category	Research Projects	Educational Use	Commercial Developments	Network and Testing
Faculty	Full Access	Full Access	Limited (Needs approval)	Allowed
Graduate Students	Full Access	Full Access	Not allowed	Allowed
Undergraduate Students	Limited	Full Access	Not allowed	Allowed
Research Staff	Full Access	Limited	Not allowed	Allowed
Visiting Researchers	Project Specific	Limited	Not allowed	Restricted Areas

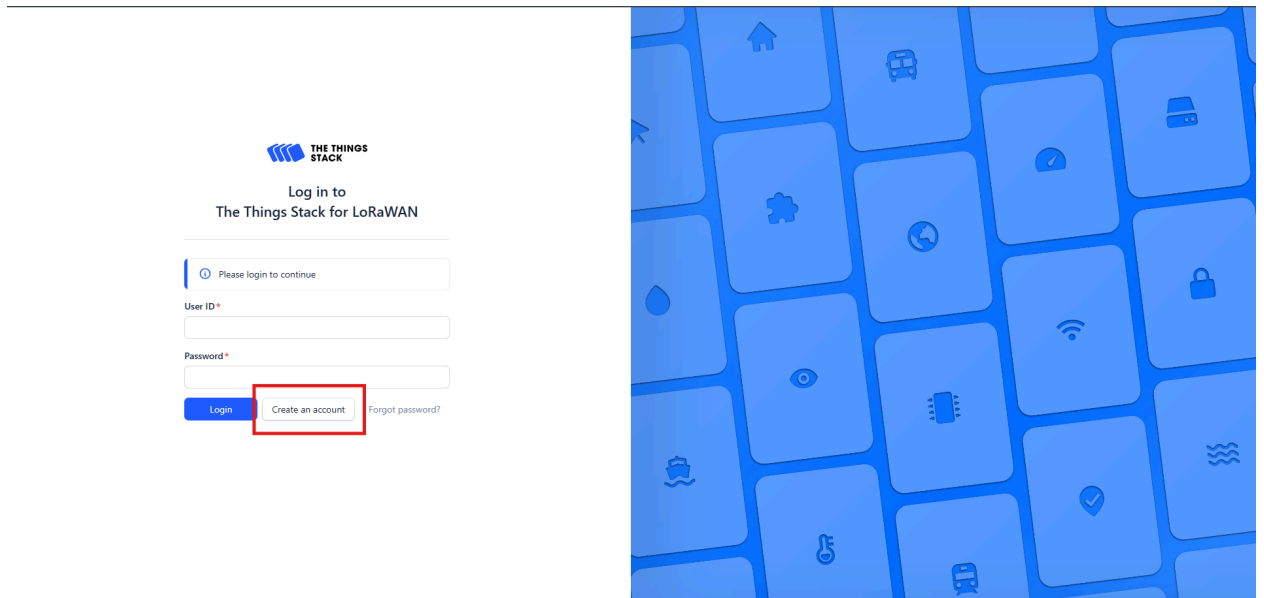
Technical Limitations

To ensure fair access and regulatory compliance, a maximum of 30 seconds of uplink airtime per device per day prevents network congestion and ensures equitable access. A maximum of 10 downlink messages per device per day conserves network resources for the broader community. Compliance with duty cycle regulations (1% maximum) ensures adherence to regulatory requirements. Adherence to regional frequency allocation rules maintains legal operation within Kenya's telecommunications framework.

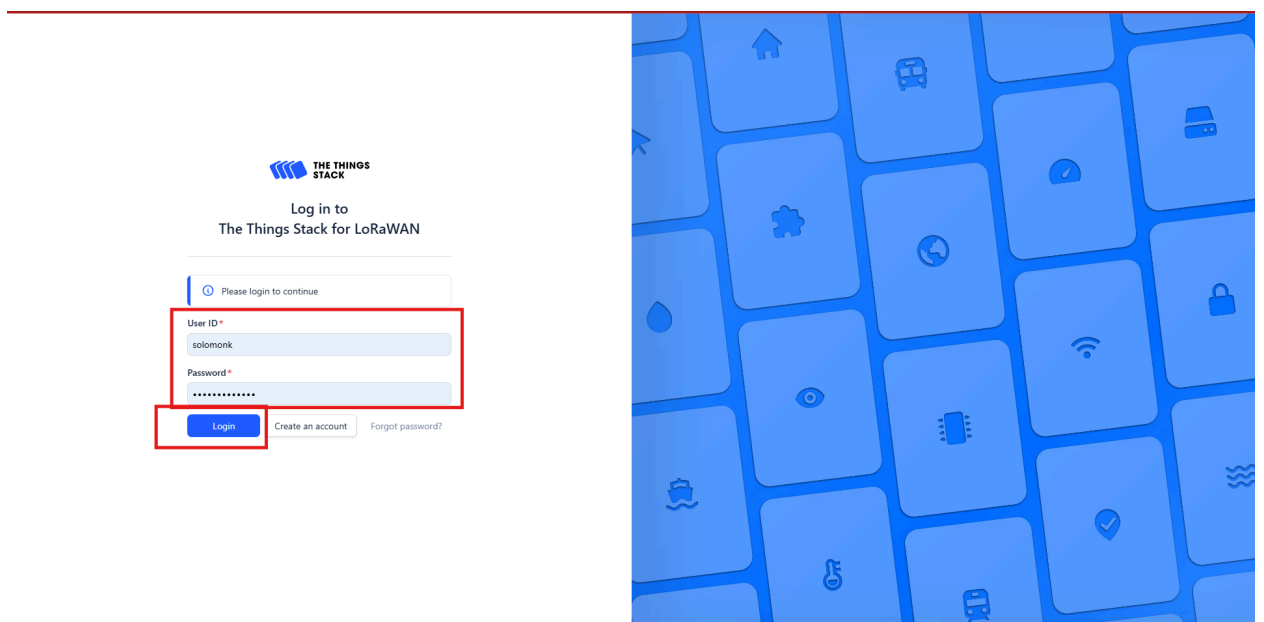
5. Getting Started: Network Access Request

- a) Create a new user account.

Use this website address: <https://tn.wirelessplanet.co.ke/console>. Click Create an account to enter the New account and password interface.

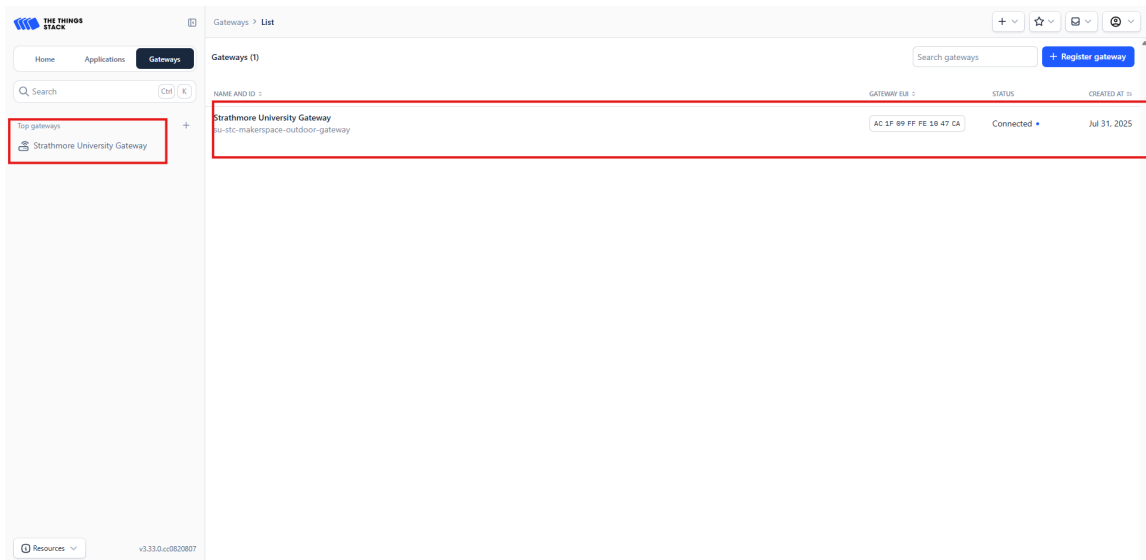
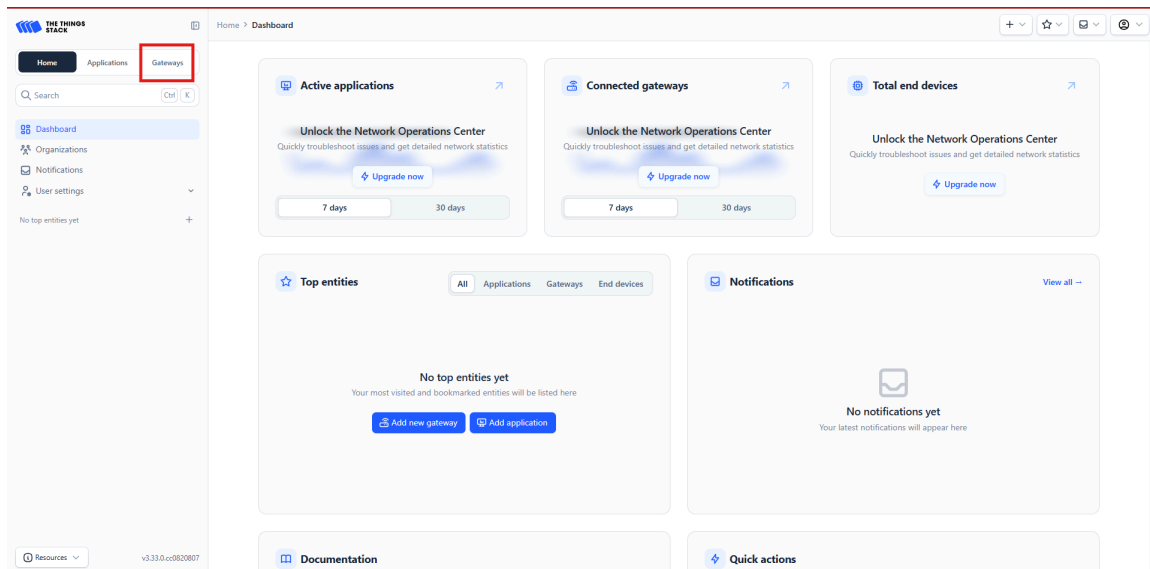


- b) Fill in this [form](#) to be granted access to the gateway
- c) Once the process is complete, we will notify you, and you can log in to your account and have access to the network



After logging in, you will be directed to the dashboard. If you have not yet been added to any Networks and would like to add yours, contact us for assistance.

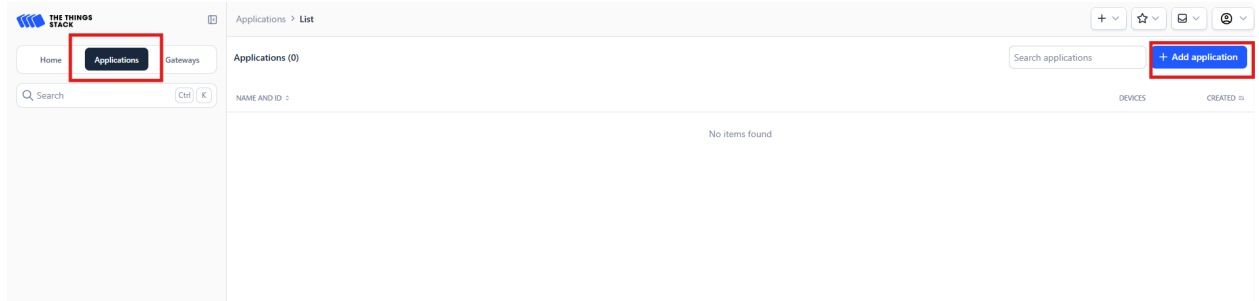
d) Navigate to see the network that you are allowed to work with



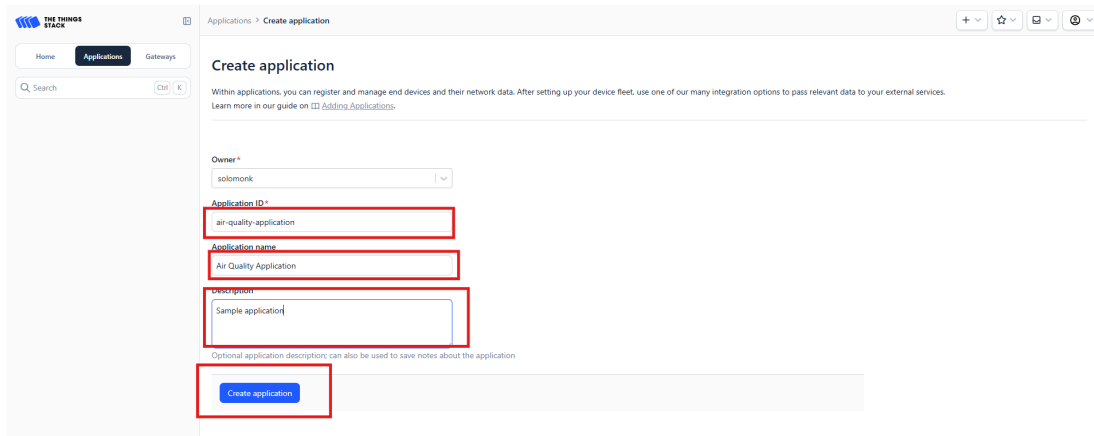
Congratulations, you have successfully joined the University LoRaWAN Network

6. Creating Application

- a) To create your first application, navigate to the application tab and click on add application button



- b) Name this application based on the project (in this case, let's use air-quality as an example) and click on the Create Application button



Congratulations, you have created your first application to manage your Air Quality devices